ASSIGNMENT 2

Exploring Text Generation Using GPT-2

1. Objective

The objective of this assignment is to understand how different text-generation parameters influence the output produced by a pretrained GPT-2 language model. The experiments demonstrate how parameters such as **temperature, top-k, top-p, and repetition penalty** affect creativity, randomness, predictability, and coherence.

2. Model and Dataset

For this assignment, the pretrained GPT-2 model from the Hugging Face Transformers library was used. GPT-2 is a language-generation model that predicts the next words based on the given input prompt.

Model used: GPT-2
Library: Hugging Face Transformers
Task: Text Generation

No separate dataset was required because the pretrained GPT-2 model was used.

3. Original Prompt

The same opening sentence was used for the five experiments:

"Artificial intelligence is changing the way we"

Only the generation parameter was changed between experiments.

Experimental Results

EXPERIMENT	PARAMETER CHANGED	VALUE	EXPECTED OBSERVATION
1	Temperature	0.5	More predictable and focused
2	Temperature	1.0	Balanced and natural
3	Top-k	20	More controlled output
4	Top-p	0.7	Focused and relatively coherent
5	Repetition Penalty	1.5	Less repetition

Experiment 1 – Temperature = 0.5



Experiment 2 — Temperature = 1.0



Experiment 3 — Top-k = 20



Experiment 4 — Top-p = 0.7

```
#Step 9: Experiment 4 — Top-p
output4 = generate_text(
    prompt,
    temperature=1.0,
    top_p=0.7
)

print(output4)
```

```
Artificial intelligence is changing the way we live. This will change our lives. It will change our lives for the better.

The future is bright and bright, but we have to learn how to live with it. We have to learn how to live with the past. We have
```

Experiment 5 — Repetition Penalty = 1.5



5. Custom Prompt

For the second part, the following custom story prompt was used:

"Once upon a time, a young girl discovered a mysterious door and it was mysterious"

Generated 3 Different Versions



6. Analysis

The experiments showed that changing the generation parameters significantly affects GPT-2's output. A lower temperature produced more predictable and focused text, while a higher temperature produced more creative and random text. Top-k and Top-p restricted the possible next words, which generally made the output more controlled. Repetition penalty helped reduce repeated words and phrases. Overall, these parameters provide control over the balance between creativity, randomness, and coherence.

7. Conclusion

GPT-2 can generate different types of text from the same prompt depending on the generation settings. Lower randomness settings are useful when we want predictable and coherent text, while higher settings can produce more creative but sometimes unusual outputs. This experiment helped demonstrate how text-generation parameters influence the behavior of a language model.